

Calculus II

Integrals of involving radicals of quadratics

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Outline

- 1 Integrals of form $\int R(x, \sqrt{ax^2 + bx + c})dx$, R - rational function
 - Transforming to the forms $\sqrt{x^2 + 1}$, $\sqrt{-x^2 + 1}$, $\sqrt{x^2 - 1}$
 - Table of Euler and trig substitutions
 - The case $\sqrt{x^2 + 1}$
 - The case $\sqrt{-x^2 + 1}$
 - The case $\sqrt{x^2 - 1}$
- 2 Rationalizing Substitutions

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Integrals of form $\int R(x, \sqrt{ax^2 + bx + c})dx$, R - rational function

Let $R(x, y)$ be an arbitrary rational expression in two variables (quotient of polynomials in two variables).

Question

Can we integrate $\int R(x, \sqrt{ax^2 + bx + c}) dx$?

- Yes. We will learn how in what follows.
- The algorithm for integration is roughly:
 - Use linear substitution to transform to one of three integrals: $\int R(x, \sqrt{x^2 + 1})dx$, $\int R(x, \sqrt{-x^2 + 1})dx$, $\int R(x, \sqrt{x^2 - 1})dx$.
 - Use trigonometric substitution or Euler substitution to transform to trigonometric or rational function integral (no radicals).
 - Solve as previously studied.
- We motivate why we need such integrals by examples such as computing the area of an ellipse.

Trigonometric Substitution

- To find the area of a circle or ellipse, one needs to compute $\int \sqrt{a^2 - x^2} dx$.
- For $\int x\sqrt{a^2 - x^2} dx$, the substitution $u = a^2 - x^2$ would work.
- For $\int \sqrt{a^2 - x^2} dx$, we need a more elaborate substitution.
- Instead, substitute $x = a \sin \theta$.

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2 \theta} = \sqrt{a^2(1 - \sin^2 \theta)} = \sqrt{a^2 \cos^2 \theta} = a|\cos \theta|.$$

- With $u = a^2 - x^2$, the new variable is a function of the old one.
- With $x = a \sin \theta$, the old variable is a function of the new one.

Linear substitutions to simplify radicals $\sqrt{ay^2 + by + c}$

- Using linear substitutions, radicals of form $\sqrt{ay^2 + by + c}$, $a \neq 0$, $b^2 - 4ac \neq 0$ can be transformed to (multiple of):
 - $\sqrt{x^2 + 1}$
 - $\sqrt{-x^2 + 1}$
 - $\sqrt{x^2 - 1}$.
- We already studied how to do that using completing the square when dealing with rational functions.

Recall: linear substitution is subst. of the form $u = px + q$.

Example

Use linear substitution to transform $\sqrt{x^2 + x + 1}$ to multiple of $\sqrt{u^2 + 1}$.

$$\begin{aligned}
 \sqrt{x^2 + x + 1} &= \sqrt{x^2 + 2 \cdot \frac{1}{2}x + \frac{1}{4} - \frac{1}{4} + 1} \\
 &= \sqrt{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} \\
 &= \sqrt{\frac{3}{4} \left(\frac{4}{3} \left(x + \frac{1}{2}\right)^2 + 1\right)} \\
 &= \frac{\sqrt{3}}{2} \sqrt{\left(\frac{2}{\sqrt{3}} \left(x + \frac{1}{2}\right)\right)^2 + 1} \\
 &= \frac{\sqrt{3}}{2} \sqrt{u^2 + 1},
 \end{aligned}$$

$$\text{where } u = \frac{2}{\sqrt{3}} \left(x + \frac{1}{2}\right) = \frac{2\sqrt{3}}{3}x + \frac{\sqrt{3}}{3}.$$

Recall: linear substitution is subst. of the form $u = px + q$.

Example

Use linear subst. to transform $\sqrt{-2x^2 + x + 1}$ to multiple of $\sqrt{-u^2 + 1}$.

$$\begin{aligned}
 \sqrt{-2x^2 + x + 1} &= \sqrt{-2\left(x^2 - \frac{1}{2}x - \frac{1}{2}\right)} \\
 &= \sqrt{-2\left(x^2 - 2\frac{1}{4}x + \frac{1}{16} - \frac{1}{16} - \frac{1}{2}\right)} \\
 &= \sqrt{-2\left(\left(x - \frac{1}{4}\right)^2 - \frac{9}{16}\right)} \\
 &= \sqrt{\frac{9}{8}\left(-\frac{16}{9}\left(x - \frac{1}{4}\right)^2 + 1\right)} \\
 &= \frac{3}{\sqrt{8}}\sqrt{-\left(\frac{4}{3}\left(x - \frac{1}{4}\right)\right)^2 + 1} \\
 &= \frac{3}{\sqrt{8}}\sqrt{-u^2 + 1},
 \end{aligned}$$

where $u = \frac{4}{3}\left(x - \frac{1}{4}\right) = \frac{4}{3}x - \frac{1}{3}$.

- Let R be a rational function in two variables.
- So far, with linear transformations we converted all integrals of the form $\int R(x, \sqrt{ax^2 + bx + c})dx$ to one of the three forms:
 $\int R(x, \sqrt{x^2 + 1})dx$, $\int R(x, \sqrt{-x^2 + 1})dx$, $\int R(x, \sqrt{x^2 - 1})dx$.
- Each of the above integrals can be transformed to a rational trigonometric integral using 3 pairs of substitutions:
 $x = \tan \theta$, $x = \cot \theta$; $x = \sin \theta$, $x = \cos \theta$; $x = \csc \theta$, $x = \sec \theta$.
- We studied that trigonometric integrals are converted to rational function integrals via $\theta = 2 \arctan t$.
- The resulting 3 pairs of substitutions are called Euler substitutions:
 $x = \tan(2 \arctan t)$, $x = \cot(2 \arctan t)$; $x = \sin(2 \arctan t)$,
 $x = \cos(2 \arctan t)$; $x = \csc(2 \arctan t)$, $x = \sec(2 \arctan t)$.
- The Euler substitutions directly transform the integral to a rational function integral.
- We will demonstrate that the Euler substitutions are rational.

Trigonometric substitution and Euler substitution

Expression	Substitution	Variable range	Relevant identity
$\sqrt{x^2 + 1}$	$x = \tan \theta$	$\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	$1 + \tan^2 \theta = \sec^2 \theta$
	$x = \cot \theta$	$\theta \in (0, \pi)$	$1 + \cot^2 \theta = \csc^2 \theta$
$\sqrt{-x^2 + 1}$	$x = \sin \theta$	$\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	$1 - \sin^2 \theta = \cos^2 \theta$
	$x = \cos \theta$	$\theta \in (0, \pi)$	$1 - \cos^2 \theta = \sin^2 \theta$
$\sqrt{x^2 - 1}$	$x = \csc \theta$	$\theta \in \left[0, \frac{\pi}{2}\right) \cup \left[\pi, \frac{3\pi}{2}\right)$	$\csc^2 \theta - 1 = \cot^2 \theta$
	$x = \sec \theta$	$\theta \in \left[0, \frac{\pi}{2}\right) \cup \left[\pi, \frac{3\pi}{2}\right)$	$\sec^2 \theta - 1 = \tan^2 \theta$

Euler substitution by applying in addition $\theta = 2 \arctan t$

$\sqrt{x^2 + 1}$	$x = \frac{2t}{1-t^2}$	$-1 < t < 1$	(?)
	$x = \frac{1}{2} \left(\frac{1}{t} - t \right)$	$0 < t$	(?)
$\sqrt{-x^2 + 1}$	$x = \frac{2t}{1+t^2}$	$-1 \leq t \leq 1$	(?)
	$x = \frac{1-t^2}{1+t^2}$	$0 < t$	(?)
$\sqrt{x^2 - 1}$	$x = \frac{1}{2} \left(\frac{1}{t} + t \right)$	$t \in (-\infty, -1) \cup [0, 1)$	(?)
	$x = \frac{1+t^2}{1-t^2}$	$t \in (-\infty, -1) \cup [0, 1)$	(?)

Trigonometric substitution $x = \cot \theta$ for $\sqrt{x^2 + 1}$

The trigonometric substitution $x = \cot \theta$, $\theta \in (0, \pi)$ for $\sqrt{x^2 + 1}$:

$$\begin{aligned}
 \sqrt{x^2 + 1} &= \sqrt{\cot^2 \theta + 1} \\
 &= \sqrt{\frac{\cos^2 \theta}{\sin^2 \theta} + 1} \\
 &= \sqrt{\frac{\cos^2 \theta + \sin^2 \theta}{\sin^2 \theta}} \\
 &= \sqrt{\frac{1}{\sin^2 \theta}} = \frac{1}{\sqrt{\sin^2 \theta}} \\
 &= \frac{1}{\sin \theta} = \csc \theta .
 \end{aligned}$$

when $\theta \in (0, \pi)$ we have
 $\sin \theta \geq 0$ and so
 $\sqrt{\sin^2 \theta} = \sin \theta$

Trigonometric substitution $x = \cot \theta$ for $\sqrt{x^2 + 1}$

The trigonometric substitution $x = \cot \theta$, $\theta \in (0, \pi)$ for $\sqrt{x^2 + 1}$:

$$\sqrt{x^2 + 1} = \frac{1}{\sin \theta} = \csc \theta .$$

The differential dx can be expressed via $d\theta$ from $x = \cot \theta$. To summarize:

Definition

The trigonometric substitution $x = \cot \theta$, $\theta \in (0, \pi)$ for $\sqrt{x^2 + 1}$ is given by:

$$\begin{aligned} x &= \cot \theta \\ \sqrt{x^2 + 1} &= \frac{1}{\sin \theta} = \csc \theta \\ dx &= -\frac{d\theta}{\sin^2 \theta} = -\csc^2 \theta d\theta \\ \theta &= \operatorname{arccot} x . \end{aligned}$$

Example

$$\begin{aligned}
 \int \frac{1}{x^2 \sqrt{x^2 + 9}} dx &= \int \frac{1}{x^2 3 \sqrt{\left(\frac{x}{3}\right)^2 + 1}} dx \\
 &= \int \frac{1}{(3 \cot \theta)^2 3 \sqrt{\cot^2 \theta + 1}} d(3 \cot \theta) \\
 &= \int \frac{1}{27 \cot^2 \theta \sqrt{\csc^2 \theta}} (-3 \csc^2 \theta) d\theta \\
 &= \frac{1}{9} \int \frac{-\csc^2 \theta}{\cot^2 \theta \csc \theta} d\theta \\
 &= \frac{1}{9} \int \frac{-\sin \theta}{\cos^2 \theta} d\theta = \frac{1}{9} \int \frac{1}{\cos^2 \theta} d(\cos \theta)
 \end{aligned}$$

Set

$$\frac{x}{3} = \cot \theta$$

$$x = 3 \cot \theta$$

$$\theta \in (0, \pi)$$

$$\theta \in (0, \pi) \Rightarrow \csc \theta > 0$$

Set $u = \cos \theta$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \cot \theta$$

$$\begin{aligned}
 &= \cot(2 \arctan t) && | \text{Recall: } \cot(2z) = \frac{\cos(2z)}{\sin(2z)} = \frac{1 - \tan^2 z}{2 \tan z} \\
 &= \frac{1 - \tan^2(\arctan t)}{2 \tan(\arctan t)} \\
 &= \frac{1 - t^2}{2t} \\
 &= \frac{1}{2} \left(\frac{1}{t} - t \right) .
 \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \frac{1}{2} \left(\frac{1}{t} - t \right) .$$

We can furthermore compute

$$\begin{aligned} \sqrt{x^2 + 1} &= \sqrt{\frac{1}{4} \left(\frac{1}{t} - t \right)^2 + 1} \\ &= \frac{1}{2} \sqrt{\left(\frac{1}{t} - t \right)^2 + 4} \quad \mid \quad \left(\frac{1}{t} - t \right)^2 + 4 = \left(\frac{1}{t} + t \right)^2 \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \frac{1}{2} \left(\frac{1}{t} - t \right) .$$

We can furthermore compute

$$\begin{aligned} \sqrt{x^2 + 1} &= \sqrt{\frac{1}{4} \left(\frac{1}{t} - t \right)^2 + 1} \\ &= \frac{1}{2} \sqrt{\left(\frac{1}{t} + t \right)^2} \quad \left| \begin{array}{l} \sqrt{\left(\frac{1}{t} + t \right)^2} = \frac{1}{t} + t \\ \text{because } t > 0 \end{array} \right. \\ &= \frac{1}{2} \left(\frac{1}{t} + t \right) . \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \frac{1}{2} \left(\frac{1}{t} - t \right) .$$

We can furthermore compute

$$\sqrt{x^2 + 1} = \frac{1}{2} \left(\frac{1}{t} + t \right) .$$

Finally compute

$$\begin{aligned} dx &= d \left(\frac{1}{2} \left(\frac{1}{t} - t \right) \right) = -\frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt \\ t &= \frac{1}{2} \left(\frac{1}{t} + t \right) - \frac{1}{2} \left(\frac{1}{t} - t \right) = \sqrt{x^2 + 1} - x . \end{aligned}$$

Euler subst. for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$

- $x = \cot \theta$ transforms $dx, x, \sqrt{x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

Definition

The Euler substitution for $\sqrt{x^2 + 1}$ corresponding to $x = \cot \theta$ is given by:

$$\begin{aligned} x &= \frac{1}{2} \left(\frac{1}{t} - t \right), & t > 0 \\ \sqrt{x^2 + 1} &= \frac{1}{2} \left(\frac{1}{t} + t \right) \\ dx &= -\frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt \\ t &= \sqrt{x^2 + 1} - x \end{aligned}$$

Euler substitution: $x = \frac{1}{2} \left(\frac{1}{t} - t \right)$, $\sqrt{x^2 + 1} = \frac{1}{2} \left(\frac{1}{t} + t \right)$,
 $t = \sqrt{x^2 + 1} - x$, $dx = -\frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt$. Recall $t > 0$.

Example

$$\begin{aligned}
 \int \sqrt{x^2 + 1} \, dx &= - \int \frac{1}{2} \left(\frac{1}{t} + t \right) \frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt \\
 &= -\frac{1}{4} \int \left(\frac{1}{t^3} + 2\frac{1}{t} + t \right) dt \\
 &= -\frac{1}{4} \left(-\frac{t^{-2}}{2} + 2 \ln |t| + \frac{t^2}{2} \right) + C \\
 &= \frac{1}{2} \left(\frac{1}{2} \left(t^{-1} - t \right) \frac{1}{2} \left(t^{-1} + t \right) \right) - \frac{1}{2} \ln t + C \\
 &= \frac{1}{2} x \sqrt{x^2 + 1} - \frac{1}{2} \ln \left(\sqrt{x^2 + 1} - x \right) + C \\
 &= \frac{1}{2} x \sqrt{x^2 + 1} + \frac{1}{2} \ln \frac{\sqrt{x^2 + 1} + x}{\left(\sqrt{x^2 + 1} - x \right) \left(\sqrt{x^2 + 1} + x \right)} + C \\
 &= \frac{1}{2} x \sqrt{x^2 + 1} + \frac{1}{2} \ln \left(\sqrt{x^2 + 1} + x \right) + C
 \end{aligned}$$

Example

Find the area locked b-n the hyperbolas $y = \pm\sqrt{x^2 + 1}$ and $x = \pm 2\sqrt{2}$.

We studied $v = \frac{1}{2} \frac{y}{u}$ is called a hyperbola: why do we call $y = \sqrt{x^2 + 1}$ hyperbola? Compute:

Signed distance b-n (x, y) and line $u = 0$ equals

$$\begin{aligned} & \pm \sqrt{\left(x - \frac{(x+y)}{2}\right)^2 + \left(y - \frac{(x+y)}{2}\right)^2} \\ &= \pm \sqrt{\frac{1}{2}(y-x)^2} = \pm \frac{\sqrt{2}}{2}(y-x) = u. \end{aligned}$$

$$\begin{aligned} \sqrt{x^2 + 1} &= y \\ x^2 + 1 &= y^2 \\ y^2 - x^2 &= 1 \\ \frac{\sqrt{2}}{2}(y-x) \frac{\sqrt{2}}{2}(y+x) &= \frac{1}{2} \\ uv &= \frac{1}{2} \\ v &= \frac{1}{2} \frac{1}{u}, \end{aligned}$$

where $\begin{cases} u = \frac{\sqrt{2}}{2}(y-x) \\ v = \frac{\sqrt{2}}{2}(y+x) \end{cases}$. Consider an arbitrary point (x, y) .

Example

Find the area locked b-n the hyperbolas $y = \pm\sqrt{x^2 + 1}$ and $x = \pm 2\sqrt{2}$.

We studied $v = \frac{1}{u}$ is called a hyperbola: why do we call $y = \sqrt{x^2 + 1}$ hyperbola? Compute:

Signed distance b-n (x, y) and line $u = 0$ equals u . Similarly compute that signed distance b-n (x, y) and the line $v = 0$ equals v .
 $\Rightarrow y^2 - x^2 = 1$ is the hyperbola $v = \frac{1/2}{u}$ in the (u, v) -plane.

$$\begin{aligned} \sqrt{x^2 + 1} &= y \\ x^2 + 1 &= y^2 \\ y^2 - x^2 &= 1 \\ \frac{\sqrt{2}}{2}(y - x) \frac{\sqrt{2}}{2}(y + x) &= \frac{1}{2} \\ uv &= \frac{1}{2} \\ v &= \frac{1/2}{u}, \end{aligned}$$

where $\begin{cases} u = \frac{\sqrt{2}}{2}(y - x) \\ v = \frac{\sqrt{2}}{2}(y + x) \end{cases}$. Consider an arbitrary point (x, y) .

Example

Find the area locked b-n the hyperbolas $y = \pm\sqrt{x^2 + 1}$ and $x = \pm 2\sqrt{2}$.
The area in question is:

$$\begin{aligned}
 & \int_{-2\sqrt{2}}^{2\sqrt{2}} 2\sqrt{x^2 + 1} dx \\
 &= 2 \left[x\sqrt{x^2 + 1} \right. \\
 & \quad \left. + \ln \left(\sqrt{x^2 + 1} + x \right) \right]_0^{2\sqrt{2}} \\
 &= 2 \left(2\sqrt{2}\sqrt{(2\sqrt{2})^2 + 1} \right. \\
 & \quad \left. + \ln \left(\sqrt{(2\sqrt{2})^2 + 1} + 2\sqrt{2} \right) \right) \\
 &= 12\sqrt{2} + 2 \ln \left(3 + 2\sqrt{2} \right) \\
 &\approx 20.496
 \end{aligned}$$

Example

Find the area locked b-n the hyperbolas $y = \pm\sqrt{x^2 + 1}$ and $x = \pm 2\sqrt{2}$.
The area in question is:

- Recall: integral can be solved via $x = \tan \theta$.
- Geometric interpretation of θ ?

$$\begin{aligned}
 & \int_{-2\sqrt{2}}^{2\sqrt{2}} 2\sqrt{x^2 + 1} dx \\
 &= 2 \left[x\sqrt{x^2 + 1} + \ln \left(\sqrt{x^2 + 1} + x \right) \right]_0^{2\sqrt{2}} \\
 &= 2 \left(2\sqrt{2}\sqrt{(2\sqrt{2})^2 + 1} + \ln \left(\sqrt{(2\sqrt{2})^2 + 1} + 2\sqrt{2} \right) \right) \\
 &= 12\sqrt{2} + 2 \ln \left(3 + 2\sqrt{2} \right) \\
 &\approx 20.496
 \end{aligned}$$

Example

Find $\int \frac{x}{\sqrt{x^2+4}} dx$.

- We could use the trig substitution $x = 2 \tan \theta$.
- But there is an easier way:
- $u = x^2 + 4$.
- $du = 2x dx$.

$$\int \frac{x}{\sqrt{x^2+4}} dx = \frac{1}{2} \int \frac{du}{\sqrt{u}} = \sqrt{u} + C = \sqrt{x^2+4} + C$$

Trigonometric substitution $x = \cos \theta$ for $\sqrt{-x^2 + 1}$

The trigonometric substitution $x = \cos \theta$, $\theta \in [0, \pi]$ for $\sqrt{-x^2 + 1}$:

$$\begin{aligned}
 \sqrt{-x^2 + 1} &= \sqrt{1 - \cos^2 \theta} \\
 &= \sqrt{\sin^2 \theta} \\
 &= \sin \theta \quad .
 \end{aligned}
 \quad \left| \begin{array}{l} \text{when } \theta \in [0, \pi] \text{ we have} \\ \sin \theta \geq 0 \text{ and so } \sqrt{\sin^2 \theta} = \sin \theta \end{array} \right.$$

To summarize:

Definition

The trigonometric substitution $x = \cos \theta$, $\theta \in [0, \pi]$ for $\sqrt{-x^2 + 1}$ is given by:

$$\begin{aligned}
 x &= \cos \theta \\
 \sqrt{-x^2 + 1} &= \sin \theta \\
 dx &= -\sin \theta d\theta \\
 \theta &= \arccos x \quad .
 \end{aligned}$$

Example

Evaluate $\int \frac{\sqrt{9-x^2}}{x^2} dx$.

- Let $x = 3 \sin \theta$, where $-\pi/2 \leq \theta \leq \pi/2$.

- Then $dx = 3 \cos \theta d\theta$.

$$\sqrt{9-x^2} = \sqrt{9-9\sin^2\theta} = \sqrt{9\cos^2\theta} = 3|\cos\theta| = 3\cos\theta$$

$$\begin{aligned} \int \frac{\sqrt{9-x^2}}{x^2} dx &= \int \frac{3\cos\theta}{9\sin^2\theta} 3\cos\theta d\theta = \int \cot^2\theta d\theta \\ &= \int (\csc^2\theta - 1) d\theta = -\cot\theta - \theta + C \\ &= -\frac{\sqrt{9-x^2}}{x} - \arcsin\left(\frac{x}{3}\right) + C \end{aligned}$$

Example

Find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a, b > 0$.

The area in question is

Express y via x :

$$\int_{-a}^a 2b\sqrt{1 - \frac{x^2}{a^2}} dx$$

$$= 4 \int_0^a b\sqrt{1 - \frac{x^2}{a^2}} dx.$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$y^2 = b^2 \left(1 - \frac{x^2}{a^2}\right)$$

$$y = \pm b\sqrt{1 - \frac{x^2}{a^2}}$$

Example

Find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a, b > 0$.

Trig subst.: set $x = a \sin \theta$, $\theta \in (0, \frac{\pi}{2})$.

Compute: $\sqrt{1 - \frac{x^2}{a^2}} = \sqrt{1 - \frac{a^2 \sin^2 \theta}{a^2}} =$

$\sqrt{1 - \sin^2 \theta} = \cos \theta$. When $x = 0$, $\theta = 0$ and when $x = a$, $\theta = \frac{\pi}{2}$.

The area in question is

$$\begin{aligned} & \int_{-a}^a 2b \sqrt{1 - \frac{x^2}{a^2}} dx \\ &= 4 \int_0^a b \sqrt{1 - \frac{x^2}{a^2}} dx \\ &= 4b \frac{a\pi}{4} = \pi ab. \end{aligned}$$

$$\begin{aligned} \int_0^a \sqrt{1 - \frac{x^2}{a^2}} dx &= \int_0^{\frac{\pi}{2}} \cos \theta d(a \sin \theta) \\ &= a \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta \\ &= a \int_0^{\frac{\pi}{2}} \frac{\cos(2\theta) + 1}{2} d\theta \\ &= a \left[\frac{\sin(2\theta)}{4} + \frac{\theta}{2} \right]_{\theta=0}^{\theta=\frac{\pi}{2}} \\ &= a \left(0 + \frac{\pi}{4} - (0 + 0) \right) \\ &= \frac{a\pi}{4} \end{aligned}$$

Example

Evaluate $\int \frac{x}{\sqrt{3-2x-x^2}} dx$.

- Complete the square under the root sign:
- $3 - 2x - x^2 = 3 + 1 - (x^2 + 2x + 1) = 4 - (x + 1)^2$
- Substitute $u = x + 1$. Then $du = dx$ and $x = u - 1$.
- $\int \frac{x}{\sqrt{3-2x-x^2}} dx = \int \frac{x}{\sqrt{4-(x+1)^2}} dx = \int \frac{u-1}{\sqrt{4-u^2}} du$
- Let $u = 2 \sin \theta$, where $-\pi/2 \leq \theta \leq \pi/2$. Then $du = 2 \cos \theta d\theta$.

$$\bullet \sqrt{4 - u^2} = \sqrt{4 - 4 \sin^2 \theta} = \sqrt{4 \cos^2 \theta} = 2 |\cos \theta| = 2 \cos \theta$$

$$\begin{aligned} \int \frac{x}{\sqrt{3-2x-x^2}} dx &= \int \frac{u-1}{\sqrt{4-u^2}} du = \int \frac{2 \sin \theta - 1}{2 \cos \theta} 2 \cos \theta d\theta \\ &= \int (2 \sin \theta - 1) d\theta = -2 \cos \theta - \theta + C \\ &= -\sqrt{4 - u^2} - \sin^{-1} \left(\frac{u}{2} \right) + C \\ &= -\sqrt{3 - 2x - x^2} - \sin^{-1} \left(\frac{x+1}{2} \right) + C \end{aligned}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$\begin{aligned}
 x &= \cos \theta \\
 &= \cos(2 \arctan t) & \left| \cos(2z) = \frac{1 - \tan^2 z}{1 + \tan^2 z} \right. \\
 &= \frac{1 - \tan^2(\arctan t)}{1 + \tan^2(\arctan t)} \\
 &= \frac{1 - t^2}{1 + t^2}
 \end{aligned}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \frac{1 - t^2}{1 + t^2}$$

$$\begin{aligned} \sqrt{-x^2 + 1} &= \sqrt{1 - \left(\frac{1 - t^2}{1 + t^2}\right)^2} \\ &= \sqrt{\frac{(1 + t^2)^2 - (1 - t^2)^2}{(1 + t^2)^2}} & | (1 + t^2)^2 - (1 - t^2)^2 = 4t^2 \\ &= \sqrt{\frac{4t^2}{(1 + t^2)^2}} & | \sqrt{4t^2} = 2t \text{ because } t > 0 \\ &= \frac{2t}{1 + t^2} \end{aligned}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

$$x = \frac{1 - t^2}{1 + t^2}$$

$$\sqrt{-x^2 + 1} = \frac{2t}{1 + t^2}$$

$$t = \frac{\sqrt{1-x} \sqrt{1+x}}{\sqrt{1+x} \sqrt{1+x}} = \frac{\sqrt{-x^2 + 1}}{x + 1}$$

we use $t > 0$

$$\begin{aligned} dx &= d\left(\frac{1 - t^2}{1 + t^2}\right) = d\left(\frac{2 - (1 + t^2)}{1 + t^2}\right) \\ &= d\left(\frac{2}{1 + t^2} - 1\right) = -\frac{4t}{(1 + t^2)^2} dt \end{aligned}$$

Euler subst. for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$

- $x = \cos \theta$ transforms $dx, x, \sqrt{-x^2 + 1}$ to trig form.
- $\theta = 2 \arctan t, t > 0$ transforms $d\theta, \cos \theta, \sin \theta$ to rational form.

What if we compose the above? We get the Euler substitution:

Definition

The Euler substitution for $\sqrt{-x^2 + 1}$ corresponding to $x = \cos \theta$ is given by:

$$\begin{aligned} x &= \frac{1 - t^2}{1 + t^2}, & t > 0 \\ \sqrt{-x^2 + 1} &= \frac{2t}{1 + t^2} \\ dx &= -\frac{4t}{(t^2 + 1)^2} dt \\ t &= \frac{\sqrt{-x^2 + 1}}{x + 1}. \end{aligned}$$

Trigonometric substitution $x = \sec \theta$ for $\sqrt{x^2 - 1}$

The trigonometric substitution $x = \sec \theta$, $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$:

$$\begin{aligned}
 \sqrt{x^2 - 1} &= \sqrt{\sec^2 \theta - 1} \\
 &= \sqrt{\frac{1}{\cos^2 \theta} - 1} \\
 &= \sqrt{\frac{\sin^2 \theta}{\cos^2 \theta}} \\
 &= \sqrt{\tan^2 \theta} \\
 &= \tan \theta .
 \end{aligned}$$

when $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$ we have
 $\tan \theta \geq 0$ and so $\sqrt{\tan^2 \theta} = \tan \theta$

Trigonometric substitution $x = \sec \theta$ for $\sqrt{x^2 - 1}$

The trigonometric substitution $x = \sec \theta$, $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$:

$$\sqrt{x^2 - 1} = \tan \theta .$$

Definition

The trigonometric substitution $x = \sec \theta$, $\theta \in (0, \pi)$ for $\sqrt{x^2 - 1}$ is given by:

$$\begin{aligned} x &= \sec \theta = \frac{1}{\cos \theta} & \theta &\in \left[0, \frac{\pi}{2}\right) \cup \left[\pi, \frac{3\pi}{2}\right) \\ \sqrt{x^2 - 1} &= \tan \theta \\ dx &= \frac{\sin \theta}{\cos^2 \theta} d\theta = \sec \theta \tan \theta d\theta \\ \theta &= \operatorname{arcsec} x . \end{aligned}$$

Example

Find $\int \frac{dx}{\sqrt{x^2 - a^2}}$, $a > 0$.

- $x = a \sec \theta$,

$$0 < \theta < \pi/2 \text{ or}$$

$$\pi < \theta < 3\pi/2.$$

- $dx = a \sec \theta \tan \theta d\theta$.

$$\sqrt{x^2 - a^2} = \sqrt{a^2 \sec^2 \theta - a^2} = \sqrt{a^2 \tan^2 \theta} = a |\tan \theta| = a \tan \theta$$

$$\begin{aligned} \int \frac{dx}{\sqrt{x^2 - a^2}} &= \int \frac{a \sec \theta \tan \theta d\theta}{a \tan \theta} = \int \sec \theta d\theta \\ &= \ln |\sec \theta + \tan \theta| + C = \ln \left| \frac{x}{a} + \frac{\sqrt{x^2 - a^2}}{a} \right| + C \\ &= \ln \left| x + \sqrt{x^2 - a^2} \right| + C_1 \end{aligned}$$

Euler substitution $x = \sec \theta$, $\theta = 2 \arctan t$

- $x = \sec \theta$ transforms dx , x , $\sqrt{x^2 - 1}$ to trig form.
- $\theta = 2 \arctan t$, $t \in (-\infty, -1) \cup [0, 1)$ rationalizes $d\theta$, $\cos \theta$, $\sin \theta$.

What if we compose the above? We get the Euler substitution:

$$\begin{aligned}
 x &= \sec \theta = \frac{1}{\cos \theta} \\
 &= \frac{1}{\cos(2 \arctan t)} \\
 &= \frac{1 + \tan^2(\arctan t)}{1 - \tan^2(\arctan t)} \\
 &= \frac{1 + t^2}{1 - t^2} = \frac{2 - (1 - t^2)}{1 - t^2} \\
 &= -1 + \frac{2}{1 - t^2}
 \end{aligned}
 \quad \left| \quad \cos(2z) = \frac{1 - \tan^2 z}{1 + \tan^2 z} \right.$$

Euler substitution $x = \sec \theta$, $\theta = 2 \arctan t$

- $x = \sec \theta$ transforms $dx, x, \sqrt{x^2 - 1}$ to trig form.
- $\theta = 2 \arctan t$, $t \in (-\infty, -1) \cup [0, 1)$ rationalizes $d\theta, \cos \theta, \sin \theta$.

What if we compose the above? We get the Euler substitution:

$$x = -1 + \frac{2}{1 - t^2}$$

$$\begin{aligned} \sqrt{x^2 - 1} &= \sqrt{\left(\frac{1 + t^2}{1 - t^2}\right)^2 - 1} \\ &= \sqrt{\frac{(1 + t^2)^2 - (1 - t^2)^2}{(1 - t^2)^2}} \quad \left| (1 + t^2)^2 - (1 - t^2)^2 = 4t^2 \right. \\ &= \sqrt{\frac{4t^2}{(1 - t^2)^2}} \quad \left| \begin{array}{l} t, 1 - t^2 \text{ have same sign} \\ \text{when } t \in (-\infty, -1) \cup [0, 1) \end{array} \right. \\ &= \frac{2t}{1 - t^2} \end{aligned}$$

Euler substitution $x = \sec \theta$, $\theta = 2 \arctan t$

- $x = \sec \theta$ transforms $dx, x, \sqrt{x^2 - 1}$ to trig form.
- $\theta = 2 \arctan t$, $t \in (-\infty, -1) \cup [0, 1)$ rationalizes $d\theta, \cos \theta, \sin \theta$.

What if we compose the above? We get the Euler substitution:

$$x = -1 + \frac{2}{1 - t^2}$$

$$\sqrt{x^2 - 1} = \frac{2t}{1 - t^2}$$

$$x = \frac{1 + t^2}{1 - t^2}$$

$$(1 - t^2)x = 1 + t^2$$

$$(1 + x)t^2 = x - 1$$

$$t^2 = \frac{x - 1}{x + 1}$$

$$t = \pm \sqrt{\frac{x - 1}{x + 1}}$$

Euler substitution $x = \sec \theta$, $\theta = 2 \arctan t$

- $x = \sec \theta$ transforms $dx, x, \sqrt{x^2 - 1}$ to trig form.
- $\theta = 2 \arctan t$, $t \in (-\infty, -1) \cup [0, 1)$ rationalizes $d\theta, \cos \theta, \sin \theta$.

What if we compose the above? We get the Euler substitution:

$$x = -1 + \frac{2}{1 - t^2}$$

$$\sqrt{x^2 - 1} = \frac{2t}{1 - t^2}$$

$$t = \pm \sqrt{\frac{x-1}{x+1}}$$

$$\begin{aligned} dx &= d\left(-1 + \frac{2}{1 - t^2}\right) \\ &= \frac{4t}{(1 - t^2)^2} dt \end{aligned}$$

Euler substitution $x = \sec \theta$, $\theta = 2 \arctan t$

- $x = \sec \theta$ transforms dx , x , $\sqrt{x^2 - 1}$ to trig form.
- $\theta = 2 \arctan t$, $t \in (-\infty, -1) \cup [0, 1)$ rationalizes $d\theta$, $\cos \theta$, $\sin \theta$.

What if we compose the above? We get the Euler substitution:

Definition

The Euler substitution for $\sqrt{x^2 - 1}$ corresponding to $x = \sec \theta$ is given by:

$$\begin{aligned} x &= \frac{1 + t^2}{1 - t^2}, & t &\in (-\infty, -1) \cup [0, 1) \\ \sqrt{x^2 - 1} &= \frac{2t}{1 - t^2} \\ dx &= \frac{4t}{(1 - t^2)^2} dt \\ t &= \pm \frac{\sqrt{x^2 - 1}}{x + 1} . \end{aligned}$$

Rationalizing Substitutions

Some non-rational fractions can be changed into rational fractions by means of appropriate substitutions. In particular, when an integrand contains an expression of the form $\sqrt[n]{g(x)}$, the substitution $u = \sqrt[n]{g(x)}$ may be effective.

Example

Let $u = \sqrt{x+4}$. Then $u^2 = x+4$, so $x = u^2 - 4$ and $dx = 2u du$.

$$\begin{aligned}\int \frac{\sqrt{x+4}}{x} dx &= \int \frac{u}{u^2-4} 2u du \\&= 2 \int \frac{u^2}{u^2-4} du \\&= 2 \int \left(1 + \frac{4}{u^2-4}\right) du && \text{long division} \\&= 2 \int du + 8 \int \frac{du}{u^2-4} \\&= 2 \int du + 8 \int \left(\frac{\frac{1}{4}}{u-2} - \frac{\frac{1}{4}}{u+2}\right) du && \text{partial fractions} \\&= 2u + 2(\ln|u-2| - \ln|u+2|) + C \\&= 2\sqrt{x+4} + 2 \ln \left| \frac{\sqrt{x+4}-2}{\sqrt{x+4}+2} \right| + C\end{aligned}$$